Reviewer Pack — Quadratic Form Rank Inverse Proportional to SOS Refutation D…

Entry ed5dcc9b878a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Quadratic Form Rank Inverse Proportional to SOS Refutation Degree for GF(2) Systems

Entry ID: `ed5dcc9b878a` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-09 19:47:16 UTC

1. Conjecture

Field A (mathematical branch): Quadratic Forms over Finite Fields **Field B** (complexity object): SOS Refutation Degree for CSPs

Statement:

> For a random system of m quadratic equations over GF(2), the SOS refutation degree required to prove unsatisfiability is Θ(√m). Specifically, for all m ≥ 2, the expected refutation degree E[δ] satisfies E[δ] = Θ(√m) with constant factors dependent on the quadratic form's rank.

Rationale (proposer's reasoning):

> Quadratic forms over finite fields encode combinatorial structures that may exhibit hidden algebraic symmetries. Their rank and rank-deficiency could influence the SOS hierarchy's ability to capture dependencies, creating a bridge between algebraic geometry and proof complexity.

Taxonomy category: `SOS_HIERARCHY` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `a94bac2e01bc0904`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM₁	LLM₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for i in range(m):
        max_row = i + max(range(i, m), key=lambda r: abs(A[r][i]))
        A[i], A[max_row] = A[max_row], A[i]
        for j in range(n):
            if i != j:
                factor = A[j][i] / A[i][i]
                for k in range(n):
                    A[j][k] -= factor * A[i][k]
    return A

def matrix_multiplication(A, B):
    m, n, p = len(A), len(B), len(B[0])
    C = [[0] * p for _ in range(m)]
    for i in range(m):
        for j in range(p):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def sos_refutation_degree(poly_system):
    m = len(poly_system)
    A = [[0] * (m + 1) for _ in range(m + 1)]
    for i in range(m):
        for j in range(i, m):
            term = poly_system[i][j]
            if term:
                A[i][j] += term
                A[j][i] += term
    A[m][m] = 1
    A = gaussian_elimination(A)
    rank = sum(1 for row in A if any(row))
    return rank
```

```python
def run_trial(seed: int) -> dict:
    random.seed(seed)
    m = 16
    instances_tested = 30
    total_ratio = 0
    counterexample = ""

    for _ in range(instances_tested):
        poly_system = [[random.choice([0, 1]) for _ in range(m)] for _ in range(m)]
        refutation_degree = sos_refutation_degree(poly_system)
        ratio = refutation_degree / math.sqrt(m)
        total_ratio += ratio

    mean_ratio = total_ratio / instances_tested
    conjecture_holds = abs(mean_ratio - 1) <= 0.2

    return {
        "metric_name": "Ratio of Refutation Degree to sqrt(m)",
        "metric_value": mean_ratio,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_ratio = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_ratio:.4f} std=0.0000 support_fraction=1.0000")
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_ratio:.4f} std=0.0000 support_fraction={support_fract
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 83, in <module>
    result = run_trial(seed)
```

```
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 62, in run_trial
    refutation_degree = sos_refutation_degree(poly_system)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 49, in sos_refutatio
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62e14fca.py", line 25, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~
ZeroDivisionError: float division by zero
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> Test crashed during Gaussian elimination with division by zero, preventing data collection. The error suggests flawed implementation of GF(2) arithmetic. | next: Fix Gaussian elimination for GF(2) by replacing division with XOR-based row operations and retest

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	97505
2	preregistration	ollama_remote	qwen3:8b	0	0	24303
3	novelty	ollama_remote	qwen3:8b	0	0	20731
4	novelty	ollama_remote	qwen3:8b	0	0	13969
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15546
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9374
7	judge	ollama_remote	qwen3:8b	0	0	20377

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 201805 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in `research/audit/ed5dcc9b878a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/ed5dcc9b878a.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/ed5dcc9b878a.tar.gz` (if generated)